ST JOSEPH'S S.S.S – NAGGALAMA

PHYSICS A'LEVEL ADAPTED CURRICULUM

SEMINAR HELD ON SATURDAY 10TH OCTOBER 2026

PAPER ONE: P510/1 (THEORY)

MARKING GUIDE

OUR ANNUAL SEMINARS ARE AS FOLLOWS;

O'LEVEL..... SECOND TERM

A'LEVEL.....THIRD TERM

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ITEM 1:

(a) Analyze the information to address the concerns of the hired worker.

Finding the volume flow rate and flow pressure of water entering the bathroom of one of the clients on the second flow of the apartments.

Volume flow rate of water entering the bathroom

Entry pipe, d_1= 2.0 cm and radius, $r_1 = \frac{2.0\times10^{-2}}{2} = 1.0 \times 10^{-2}$m, Velocity of water at the entry pipe, $V_1 = 2.0\ ms^{-1}$.

Bathroom pipe, $d_2 = 1.0\ cm$, radius, $r_2 = \frac{1.0\times10^{-2}}{2} = 5.0 \times\ 10^{-3}m$, velocity of water at bathroom pipeV_2 =?

From the equation of continuity, $AV = constant(k)$, finding velocity of water in the bathroom pipe.

$$A_1V_1 = A_2V_2 \Longrightarrow V_2 = \frac{A_1V_1}{A_2} \Longrightarrow = \frac{\pi(1.0\times10^{-2})^2\times2}{\pi(5.0\times10^{-3})^2} = 4\times2 = 8ms^{-1}$$

Volume flow rate

For an incompressible fluid the volume flow rate is the same at every cross section of the pipe.

$Q = \frac{V}{t}, Q = \frac{Al}{t}$ But $\frac{l}{t} = V \Longrightarrow$ Thus; $Q = AV$ therefore, $Q = A_1V_1$

$$Q = \pi(1.0\times10^{-2})^2 \times 2.0 = 6.28\times10^{-4}ms^{-1}$$
$$= \left((6.28\times10^{-4})\times1000\right)litres\ s^{-1} = 0.628\ litres\ s^{-1}$$

Flow pressure of the water entering the bathroom pipe

From, Bernoulli's principle; $P\ + \frac{1}{2}\rho v + \rho gh = constant\ (k)$.

$P_1 = 4.0\times10^{-5}pa, V_1 = 2.0ms^{-1}, V_2 = 8.0ms^{-1}, h_2 - h_1 = 5.0m$ and $\rho = 1000kgm^{-3}$

Then; $P_1\ + \frac{1}{2}\rho V_1^2 + \rho gh_1\ =\ P_2\ + \frac{1}{2}\rho V_2^2 + \rho gh_2$

$$P_2 = P_1 + \frac{1}{2}\rho(V_1^2 - V_2^2) - \rho g(h_2 - h_1)$$

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$$P_2 = 4.0 \times 10^5 + \frac{1}{2}1000(2.0^2 - 8.0^2) - (1000)(9.81)(5.0)$$

$$P_2 = 4.0 \times 10^5 - 3.0 \times 10^4 - 4.905 \times 10^4$$

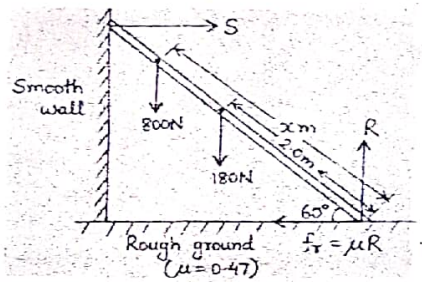
$$P_2 = 3.21 \times 10^5 Pa$$

Since the flow pressure of $3.21 \times 10^5 Pa$ is not at least $3.5 \times 10^5 Pa$ and volume flow rate of $0.628\ litres\ s^{-1}$ is not at least 0.7 litres per second. Therefore, the water supply system to the bathroom is not good.

Finding how far the worker could climb just before the ladder could slip.

Weight of the ladder= $180N$, length, $l = 4.0m$, inclined at $= 60^0$, $\mu = 0.47$, worker's weight $= 800N$.

Let, the distance along the ladder from its foot when it's about to slip be, x.



At equilibrium.

Resolving vertically,

$R = (800 + 180)N = 980N$

Resolving Horizontally,

$f_r = S$ but $f_r = \mu R$

$f_r = 0.47 \times 980 = 460.6N$

Therefore, $S = 460.6N$

Taking moments about the foot of the ladder.

From the principle of moments;

Sum of clockwise moments = sum of anticlockwise moments

$$(lsin60^0)\,S = 180\left(\frac{l}{2}cos60^0\right) + 800\,(xcos60^0)$$

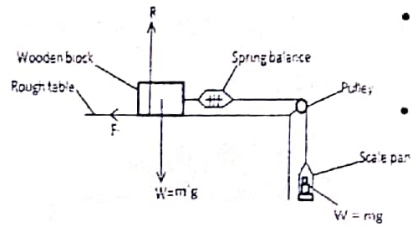
$$(4.0sin60^0)\,(460.6) = 180(2.0\,cos60^0) + 800\,(xcos60^0)$$

$$1595.565 = 180 + 400x$$

$$1415.565 = 400x \quad \Rightarrow x = 3.539m$$

The worker could climb 3.539m from the foot of the ladder just before the ladder could slip.

How to determine the coefficient of friction of a rough ground

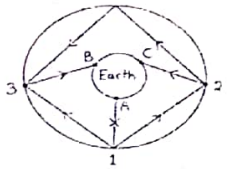


- The normal reaction is varied by adding more masses on the block.
- Masses are added on the scale pan and each time the block is given a slight push until a point when the block moves with a constant speed.
- For each mass of the block, the frictional force, F_f is determined from the fact that F_f = weight on the scale pan.
- The procedures are repeated for each mass of the block.
- The results are tabulated including the values of normal reaction R
- A graph of F_F against R is plotted and a straight line through the origin is obtained.
- The slope, S of the graph is determined.
- The coefficient due to kinetic friction, μ is then obtained from the expression;- $\mu = S$

How communication satellites distribute signals globally.

Communication satellites distribute signals using radio waves and micro waves that travel through the vacuum of space at the speed of light from one place to another as follows;

A set of at least three communication satellites is launched into a parking orbit.



Signals from the ground station are sent up wards to the satellite (1). Satellite (1) receives the signals and amplifies them. It retransmits the signals onto satellite (2) or (3).

The signals are then sent downwards to the receivers at any point on the earth's surface.

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(b) Help the plumber understand road section designs he can develop to enable him travel at highest safe speeds.

Banking the circular road so that centripetal force is provided by the horizontal component of normal reaction and friction. This provides enough force that can balance maximum centripetal force thus driving at a high speed with minimal tendency to side slip.

Increasing the radius of the circular path which reduces the centripetal force for easier negotiation at high speed without skidding.

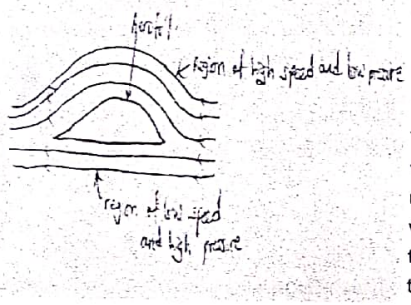
Increasing the coefficient of friction by tarmacking the road with an appropriate aggregate so as to drive safely at high speed both at the flat straight road and circular road sections.

ITEM 2:

(a) Respond to the concerns of the pilot, the rescue team and establish whether the launch of new communication satellite will be successful.

How the shape of the aircraft wings creates a lift force

The wings of the aircraft form an *aero foil lift*



An aircraft wing is shaped so that air flows faster along the top of the wing than below the wing. By Bernoulli's principle, pressure below becomes greater than that above the wings.

This pressure difference produces the resultant force called lift upward force. It is this force which provides a force that lifts the plane off the ground at takeoff.

Finding lift force of the aircraft

From, Bernoulli's principle; $P + \frac{1}{2}\rho v + \rho g h = constant\ (k)$

Then; $P_1 + \frac{1}{2}\rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g h_2$. Since the velocities of air act at the same height at the wing of the plane, thus $h_1 = h_2$.

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$

$$P_1 - P_2 = \frac{1}{2}\rho(V_1^2 - V_2^2) \text{ And from } F = PA$$

$$\text{Then, } (P_1 - P_2)A = \frac{1}{2}\rho A(V_1^2 - V_2^2) \text{ and lift force, } F = (P_1 - P_2)$$

$V_2 = 120\ ms^{-1}, V_1 = 100 ms^{-1}, \rho = 1.29 kgm^{-3}, A = 40m^2$

$$\text{Lift force} = \frac{1}{2}(1.29)(40)(120^2 - 100^2)$$

$$\text{Lift force} = 25.8\ (4400)$$

$$\text{Lift force} = 113520N$$

Total weight of the plane = $mass \times acceleration\ due\ to\ gravity$

Total weight of the plane= 1500×9.81

Total weight of the plane = $14715N$

Since the lift of 113520N of the aircraft exceeds the total weight 14715N of the plane, therefore the speeds are sufficient to cause a lift.

Why Musa felt weightless and other passengers felt a normal weight

Weight felt is due to normal reaction of whatever supports the passengers. For passengers moving downwards, acceleration, a is experienced and by Newton's Second law (downward positive) gives $mg - R = ma$ and $R = m(g - a)$

Musa's puncture removed most of the parachute's air resistance, he felt almost freely with a=g. then$R \approx m(0) \approx 0$, therefore nothing was pushing him up. So, he felt weightless (his true weight (mg) is unchanged he simply had no support force.

The other passengers felt a normal weight because their parachutes' air resistance grew until it balanced the weight and they descended at a terminal

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(constant) speed. So, $a = 0$ and $R = m(g - 0) = mg$, they felt their normal weight.

Why Musa bent his knees but others did not

Upon reaching the ground, the passenger's momentum was removed. By the impulse momentum relation, the average force on the passenger's body is;

$$F_{av} = \frac{m(V - 0)}{\Delta t}$$

Musa reached the ground with a larger speed, V or free-fall leading to a large change in momentum. Bending the knees made Musa's body to come to rest over a larger time (Δt), which greatly reduced the average force on his legs.

Other passengers landed with a small terminal speed thus their momentum to be removed was small and the force was already safe for landing hence they did not need to bend their knees.

Evaluating the success of the mission

The launch energy needed is equal to satellite's total mechanical energy, E from rest on the ground to a circular orbit.

At surface the total energy (potential energy) $= \frac{-GMm}{R}$

In the circular orbit, total energy $= \frac{-GMm}{2r}$

Increase in mechanical energy, $E = -\frac{GMm}{2r} - \frac{-GMm}{R} = -\frac{GMm}{2r} + \frac{GMm}{R}$

$E = GMm(\frac{1}{R} - \frac{1}{2r})$ and $GM = gR^2 \Rightarrow E = (gR^2)m(\frac{1}{R} - \frac{1}{2r}) = mgR(1 - \frac{R}{2r})$

$m = 2000kg$, $g = 9.81\ ms^{-2}$, $R = 6.4 \times 10^6 m$, $r = R + h$, $r = (6.4 \times 10^6) + (8.0 \times 10^5) = 7.2 \times 10^6 m$ and $h = 800km = 800000m$.

$$E = 2000 \times 9.81 \times (6.4 \times 10^6)(1 - \frac{6.4 \times 10^6}{2(7.2 \times 10^6)}) = 6.98 \times 10^{10} J$$

The mission will not succeed since the launch energy of 6.98×10^{10}J is greater than the rocket's capacity of 6.5×10^{10}J.

(b) Guide Dr. Sarah to write launching procedures, circumstances under which launching may fail and what is likely to happen if it fails and the necessary estimates.

The launching procedures

- Site and timing, a launch site is chosen near the equator, with clear weather and a clear flight path, and a launch window that avoids debris and other satellites.
- Pre-launch checks, the satellite's power, communications, antenna and solar-panel deployment systems, fuel, guidance computers and the rocket stages are tested; fuel the rocket.
- Lift-off and vertical ascent, the first stage must produce thrust greater than the total weight so the rocket rises through the dense lower atmosphere.
- Pitch-over (gravity turn), the rocket tilts gradually toward the horizontal (toward the east) so that it gains horizontal speed while climbing out of the atmosphere.
- Staging, each stage is dropped when its fuel is spent to reduce mass. The fairing is released once above the atmosphere.
- Reaching the parking orbit, the rocket coasts or burns to the required height.
- At the top the final burn gives it a horizontal speed of $v = \sqrt{\frac{GM}{r}}$, so that gravity alone provides the centripetal force. This circularizes the orbit.

Release and commissioning, the satellite separates, opens its solar panels and antennas, and ground controllers track it, test the transponders and make small corrections.

Circumstances under which launching may fail

- ✓ Satellite not having enough thrust, the satellite having thrust which is less than weight of the satellite at lift-off fails to be launched.
- ✓ Small horizontal speed at the parking orbit, since the path is a flattened ellipse that intersects the Earth, and the satellite falls back.

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- ✓ Collision with space debris or another satellite that may lead to burning out of a satellite in space due to high friction that increases the surface temperature of the satellite.
- ✓ Wrong direction or guidance fault, an orbit with the wrong inclination or a collision course.
- ✓ Engine or structural failure, vibration, fuel leak, premature stage separation or bad weather

ITEM 3:

(a) Analyze the information and educate the factory workers about their concerns

Why a beta source is preferred for the thickness gauge

The beta source is preferred because it has an ideal penetration power suitable for thin materials like aluminum foil.

How a nuclear fission reactor releases energy.

- The reactor contains fuel rods made of Uranium – 235 or plutonium – 239.
- When a neutron hits a Uranium nucleus, the nucleus becomes unstable and splits into two smaller nuclei. This is nuclear fission.
- This fission reaction releases heat energy, more neutrons, and radiation.
- The released neutrons hit other Uranium nuclei, causing more fission reactions. This is called a chain reaction.
- Control rods absorb excess neutrons to regulate the speed of the chain reaction and prevent overheating.
- The heat generated is absorbed by a coolant, which transfers it to water, producing steam.
- The steam drives a turbine, which spins a generator to produce electricity.
- The steam is then cooled back into water in a condenser and then reused.

How the reactor is controlled to keep it safe

- Using control rods like boron, cadmium to absorb neutrons thus lowering them into the core slows the reaction; raising them speeds it up; fully inserted they shut the reactor down.
- Using moderators like graphite, ordinary or heavy water to slow the fast neutrons by collisions to thermal speed, so that they are far more likely to be captured by uranium-235.
- Using coolants like water, carbon dioxide gas or liquid sodium that carry the heat from the core to the heat exchanger or boiler that drives the turbines, and prevents overheating or melt-down.
- Shielding and containment with a thick concrete and steel absorb neutrons and gamma rays and contain any release of radioactive material.
- Monitoring of neutron-flux, temperature and pressure detectors that trigger automatic shut-down; spent fuel is stored safely.

Why a neutron is used and not an alpha particle or a proton

Neutrons carry no charge and hence they are not repelled by the nucleus, unlike alpha particles or a proton that are positively charged.

Whether 1 kg of pure uranium is enough to cater for the daily deficit energy.

$$\Delta m = (235.0439 + 1.0089 - (140.9144 + 91.9262 + 3.0261)$$

$$\Delta m = (236.0526 - 235.8667)u = 0.1859u$$

$$E = 0.1859 \times 931 = 173MeV = (173.07 \times 10^{6})(1.6 \times 10^{-19})$$

$$= 2.77 \times 10^{-13}\,J$$

Then, $N = (m / M) \times N_A = (1000 / 235) \times 6.02 \times 10^{23}$

$N = 2.56 \times 10^{24}$ atoms

Calculation of Total Energy (E_{total}); $E_{total} = N \times E = (2.56 \times 10^{24}) \times (2.77 \times 10^{-11})$ J

$= 7.09 \times 10^{13}$ J

Calculation of Mass Needed;

Fraction needed = $(6.9 \times 10^{13}) / (7.09 \times 10^{13}) = 0.973$ kg

Mass needed = 0.973 kg

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Since the mass of 0.973 kg needed is less than 1 kg of pure Uranium-235, there is enough energy

(b) Help the students assess the quality of the factory products and how it monitors the thickness

Assessing the quality of the factory products

Constructive reflection occurs when the Bragg condition is satisfied,

$2d \sin\theta = n\lambda_{min}$

Finding the minimum wavelength (λ_{min})

$eV = hf$, but $f = c/\lambda_{min}$

$eV = (h\,c)/\lambda_{min}$

$\lambda_{min} = (h\,c)/(e\,V)$

$\lambda_{min} = [(6.6 \times 10^{-34}) \times (3.0 \times 10^{8})] / [(1.6 \times 10^{-19}) \times (4.08 \times 10^{4})]$

$\lambda_{min} = 3.09 \times 10^{-11}$ m

For the second order, $n = 2$, with $d = 2.3 \times 10^{-10}$ m.

$\sin\theta = (n \times \lambda)/(2d)$

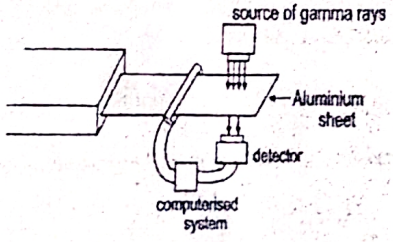
$\sin\theta = (2 \times 3.09 \times 10^{-11}) / (2 \times 2.3 \times 10^{-10}) = 0.1345$

$\theta = \sin^{-1}(0.1345)$

$\theta = 7.73^{\circ}$

Since the diffracting angle 7.73° for x-rays does not fall below 7° for the second order diffraction, therefore the aluminum sample is of high quality.

Procedure by which the factory keeps the sheet thickness uniform



- Gamma rays from the source are directed to the aluminum sheet below which is placed a detector.
- The detector reading is set at a particular value for standard thickness.

- When thickness reduces, more gamma rays pass through the sheet, the detector reading increases and signals are sent to the computerized system that commands the rollers to widen the gap.
- When thickness increases, fewer gamma rays pass through the sheet, detector reading reduces and signals are sent to the computerized system that commands rollers to narrow the gap.

ITEM 4:

(a) Address the needs of the students' group and the first instructor.

How the beam is produced and its velocity

The cathode is heated and emits electrons by thermionic emission. The electrons are accelerated towards the anode by the p.d of 18.22 kV, so the electric potential energy lost becomes kinetic energy as electrons reach the plates as a narrow beam through the holes in the anode.

Velocity of the Beam ; $eV_a = \frac{1}{2} m v^2 \Longrightarrow v = \sqrt{[(2\,e\,V_a)/m]}$

$v = \sqrt{[(2 \times (1.6 \times 10^{-19}) \times (1.822 \times 10^{4}))/(9.11 \times 10^{-31})]} = 8.0 \times 10^{7}\ m\ s^{-1}$

The Electric and Magnetic Forces on the Beam between the Plates

For electric force: From: $E = V/d = 200/(5.0 \times 10^{-3}) = 4.0 \times 10^{4}\ V\ m^{-1}$ And $F_E = eE$

Then: $F_E = (1.6 \times 10^{-19}) \times (4.0 \times 10^{4}\ V\ m^{-1}) = 6.4 \times 10^{-15}$ N

The electric force is 6.4×10^{-15} N

Magnetic Force: From; $F_B = B\,e\,v = (0.001) \times (1.6 \times 10^{-19}) \times (8.0 \times 10^{7})$

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$= 1.28 \times 10^{-14}$ N

Since the electric force, F_E, is not equal to the magnetic force, F_B, therefore the beam is deflected as it passes through the two forces of the electric and magnetic field.

Length of the Beam on the Screen

Between the plates, the electrons move at a constant speed along the plate and accelerate sideways.

From $F = ma$, then $a = F/m$, the length of the plates is 2.5 cm = 2.5 / 100 = 2.5×10^{-2} m.

Acceleration of Electrons; $a = (6.4 \times 10^{-15}) / (9.11 \times 10^{-31}) = 7.03 \times 10^{15}$ m s^{-2}

Time, $t = L / v = (2.5 \times 10^{-2}) / (8.0 \times 10^{7}) = 3.125 \times 10^{-10}$ s

Vertical Displacement of the Electrons as they leave the Plates

$$y_1 = \tfrac{1}{2} a t^2 = \tfrac{1}{2} (7.03 \times 10^{15}) (3.125 \times 10^{-10})^2$$

$$y_1 = 3.43 \times 10^{-4} \text{ m} = 0.343 \text{ mm}$$

Since 0.343 mm is less than half the plate gap of 2.5 mm, the beam passes between the plates without striking them.

Deflection beyond the plates

After leaving the plates, the beam travels in a straight line that appears to come from the middle of the plates, by similar triangles:

D = 0.40 m, l = 0.025 m

From, $Y / y_1 = D / (l/2)$

$Y = y_1 \times (D / (l/2))$

$Y = 3.43 \times 10^{-4} \times (0.40 / 0.0125)$

$Y = 1.10 \times 10^{-2}$ m = 1.10 cm

V = 200 V, d = 0.005 m, V_a = 18.22 kV = 18220 V

From, $Y = (V \cdot L \cdot D) / (2 \cdot d \cdot V_a)$

$Y = ((200) \times (0.025) \times (0.40)) / (2 \times 0.005 \times 18220) = 1.10 \times 10^{-2}$ m = 1.10 cm

Since the beam length of 1.10 cm is not at least 1.5 cm, therefore the Y-sensitivity of the C.R.O. was not met.

Needs of the First Instructor

Finding the r.m.s voltage

The instructor adjusted Y-amplifier so that 1 cm of vertical deflection represents 10 V and peak to peak height is 4 cm.

$V_{pp} = (4 \times 10) = 40$ V

Since, $V_o = V_{pp} / 2 = 40 / 2 = 20$ V

Then, $V_{rms} = V_o / \sqrt{2} = 20 / \sqrt{2} = 14.1$ V

Finding the frequency

From, f = 1 / T

T = (time (t)) / (number of complete cycles) = 0.01 / 2 = 5.0×10^{-3} s

Then, f = 1 / (5.0×10^{-3}) = 200 Hz

Why the Time Base is Necessary

- The time base sweeps the beam (spot moving up and down) across the screen at a constant speed from left to right, then returns it quickly backwards. The horizontal position then represents time and the input is displayed as a graph of voltage against time.
- For a steady trace, the time base period must be a whole multiple of signal's period that triggers each sweep to start at the same point of the wave. This horizontal scale (time per cm) allows the period and frequency to be calculated from the trace.

How the Radiation was produced

- The stream of fast-moving electrons (cathode rays) stroked the metal target embedded in the copper anode and are suddenly slowed or stopped by electrostatic forces of positive target nuclei, as 99% of the
- kinetic energy of the cathode rays is converted into heat which is conducted and taken away by cooling fins. 1% of the kinetic energy of cathode rays is used to produce X-rays in two ways:

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- Continuous X rays, are produced when a decelerating electron radiates its lost kinetic energy as an electromagnetic photon, giving a continuous range of wavelengths.
- Characteristic X rays, are produced when an incident electron knocks an inner-shell electron out of the target atom, an outer electron falls into the vacancy and emits an X ray photon of fixed energy that is characteristic of the metal target.

Why the Leaf of the Electroscope Changes Divergence

- The negatively charged gold leaf electroscope has excess electrons on the plate and leaf which repel one another and make the leaf diverge.
- When radiation in the form of X-ray photons of energy strike the metal plate, the electrons absorb this energy which is far greater than the work function of the metal. The electrons are released from the metal with maximum kinetic energy, forming a photoelectric emission.
- The plate loses negative charge, which creates a fall in repulsion between the metal plate and leaf, and the divergence of the leaf decreases. The more the intensity of the radiation, the more electrons are emitted per second and the faster the leaf collapses.

(b) Establish the operation of the system raised by the second instructor to help banks.

- The effect on the metal plate can be used to set a burglar alarm in banks for security to protect the bank from robbers.
- The beam of radiation is made to fall on a photocell or a photosensitive metal constantly across a doorway or vault entrance.
- While the beam falls on the cathode, photoelectrons are emitted and a steady photocurrent flows in the circuit, the relay keeps the alarm off.
- When a robber crosses the beam and blocks the radiation, no photoelectrons are emitted and the photocurrent drops to zero. This change switches the relay that closes the alarm circuit and the alarm sounds.

ITEM 5:

(a) Provide the necessary assistance required by the hospital safety officer

Wavelength of the radiation needed

Work function, $W_o = hf_o \Rightarrow W_o = hc/\lambda_o$

$$\lambda_o = (6.6 \times 10^{-34} \times 3.0 \times 10^8) / (1.0 \times 1.6 \times 10^{-19}) = 1.24 \times 10^{-6}\ m$$

The wavelength of the radiation which should be incident on the cell is 1.24×10^{-6} *m*

How the system works

- ✓ The 1.5V battery makes the anode A positive relative to the cathode K so that any electrons emitted from K are collected by A.
- ✓ When there is no intruder, electrons are emitted, photocurrent flows through the resistor and the p.d. across the resistor is high. The input of the NOT gate is therefore High (1) and the output of the alarm is Low (0) and the alarm is off.
- ✓ When the intruder blocks the radiations, no photoelectrons are emitted, thus no photocurrent and the p.d. across the resistor falls to zero. The NOT gate input becomes Low (0) and the output is High (1), the alarm is triggered.

Safety of X-ray machine and the intensity-wavelength graph

Total Power Supplied = 495 / 0.99 = 500 W.

X-ray Power = 0.01 × 500 = 5.0 W.

From, P = IV then, $V = P/I \Longrightarrow = 500 / (10 \times 10^{-3}) = 5.0 \times 10^4\ V = 50\ kV$

From, $eV = hf \Rightarrow f_{max} = (eV)/h = (1.6 \times 10^{-19} \times 50 \times 10^3) / (6.6 \times 10^{-34})$

$= 1.21 \times 10^{19}$ Hz

Since the frequency of 1.21×10^{19} *Hz is in the range* 8.0×10^{15} *Hz to* 3.0×10^{19} *Hz, the X-radiations produced was safe.*

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How the binding-energy-per-nucleon curve accounts for the energy released

- The curve of binding energy per nucleon against mass number rises steeply for light nuclei, reaches a maximum of about 8.8 MeV at iron-56 ($A \approx 56$) and then falls slowly for heavy nuclei.
- Hydrogen-2 (deuterium) is at the steep, low part of the curve (about 1.1 MeV per nucleon) while helium-4 lies much higher (about 7.1 MeV per nucleon).
- When two deuterium nuclei fuse, the product nucleus has a larger binding energy per nucleon, so it is more tightly bound and has lower energy. The difference is released: total binding energy increase $\approx 4 \times (7.1 - 1.1) \approx 24$ MeV per reaction.
- In terms of mass: the mass of the helium nucleus is less than the mass of the two deuterium nuclei (mass defect), and Δmc^2 is the energy released.
- Fusion requires extremely high temperature (of order 10^8 K) so that the nuclei have enough kinetic energy to overcome their mutual electrostatic repulsion, and the hot plasma must be confined (for example magnetically).

Number of reactions

Energy per reaction; $\Delta m = 2\,(2.0136) - 4.0015 = 0.0257$ u

$$E = \Delta m\, c^2 = (0.0257)\,(1.66 \times 10^{-27})\,(3.0 \times 10^8)^2 = 3.84 \times 10^{-12}\ \text{J} \approx 24.0\ \text{MeV}$$

Number of reactions and mass of hydrogen-2 for 1 MJ

$$\text{Number of reactions} = (1.0 \times 10^6) / (3.84 \times 10^{-12}) = 2.60 \times 10^{17}$$

Each reaction uses two nuclei of hydrogen-2:

$$\text{Number of nuclei} = 2 \times 2.60 \times 10^{17} = 5.21 \times 10^{17}$$

$$m = 5.21 \times 10^{17} \times 2.0136 \times 1.66 \times 10^{-27} = 1.74 \times 10^{-9}\ \text{kg}$$

Therefore, 2.6×10^{17} reactions are needed, using 1.74×10^{-9} kg (about 1.7 μg) of hydrogen-2 to produce 1 MJ.

(a) Address the concerns of the Nurse.

When Can the Solution be injected?

$A_0 = 6.0 \times 10^5$ Bq, $t_{1/2} = 8$ days

Number of atoms, $N = 5.73 \times 10^{11}$ atoms.

$$\text{From, } \lambda = \ln 2 / t_{1/2} = 0.693 / (8 \times 24 \times 3600) = 1.003 \times 10^{-6}\ s^{-1}$$

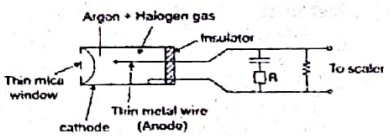
$$N_0 = A_0 / \lambda = (6.0 \times 10^5) / (1.003 \times 10^{-6}) = 5.98 \times 10^{11}\ \text{atoms}$$

From, $N = N_0 e^{-\lambda t} \Rightarrow t = (1/\lambda) \ln (N_0 / N) = [1 / (1.003 \times 10^{-6})] \ln [(5.983 \times 10^{11}) / (5.73 \times 10^{11})] = 4.31 \times 10^4$ s ≈ 12.0 hours.

The nurse must wait 12 hours

Why Iodine-131 is used Instead of X-rays:

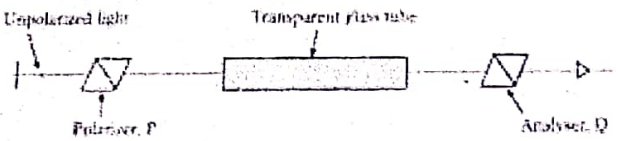
- An X-ray image shows the structure (shadow of denser tissue) therefore, it doesn't show how the gland is functioning. The thyroid gland absorbs and concentrates Iodine to make its hormone, so radioactive Iodine collects in the gland and the amount taken up reveals how the gland is working.
- Iodine 131 emits gamma rays which pass out of the neck and are detected by the probe since it has a half-life which is long enough for the test but short enough that radioactivity disappears quickly from the patient.
- **The device is a Geiger Muller tube**



  - When ionizing radiations (Iodine 131) enter the GM tube through the thin mica window, argon atoms are ionized.
  - The electrons move very fast to the anode and positive ions drift to the cathode.
- When the electrons reach the anode, a discharge occurs and a current flow in the external circuit.
- A p.d is obtained across a large resistance R which is amplified and passed to a scale.
- The magnitude of the pulse registered gives the extent to which ionization occurred.

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ITEM 6:



- ✓ A beam of un-polarized light is directed into an empty glass tube through a polarizer P.
- ✓ The emerging light is viewed through an analyzer Polaroid, the analyzer is slowly rotated in a direction perpendicular to the tube until light is just extinguished.
- ✓ The position of the analyzer at which the light is cut off is noted, sugar solution under test is then put into the glass tube. '
- ✓ Light is seen through the analyzer Polaroid. The analyzer is further rotated until the observed light is extinguished
- ✓ The new position of the analyzer is noted. The angle through which the analyzer has been rotated, θ is measured and noted
- ✓ The concentration of sugar, c is proportional to θ i.e. c=k θ, knowing the value of k, the value of c is obtained.

Wien's displacement constant, $K=\lambda_{min}T$

$$\rightarrow K = 1.07X10^{6}x10^{-9}\ x\ 2.7 = 2.889x10^{-3}\ mK$$

Surface temperature, $T_s = \frac{K}{\lambda_{min}} = \frac{2.889x10^{-3}}{500x10^{-9}} = 5778\ K$

Solar constant, $S=\frac{4\pi R_S^2}{4\pi r^2}\delta T^4 = (\frac{R_s}{r})^2\delta T^4 = \left(\frac{6.96x10^8}{1.5x10^{11}}\right)x5.67x10^{-8}x\ 5778^4 = 1360.6\ Wm^{-2}$

Power absorbed by the blackened roof is equal to the power received;

$SA = \delta AT^4$

$$\rightarrow T^4 = \frac{S}{\delta} = \frac{1360.6}{5.67X10^{-8}} = 393.6\ K$$

Since, $T< 900K$, *the roof will not melt.*

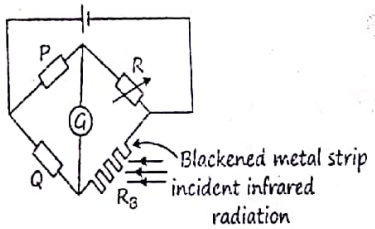
$$\frac{Q}{t} = \frac{KA(396-\theta)}{d}$$

$$S = \frac{K(393.6-\theta)}{d} \Rightarrow \theta - 393.6 = \frac{1360.6\ x\ 25\ x10^{-3}}{0.85} = 353.6K$$

$$\theta = 353.6 - 273 = 80.6^0\ C$$

Since θ *is slightly above* 80^0C, *it doesn't light between the* 50^0C *and* 80^0C, *the nutrients maybe lost.*

(b) The surface temperature of the roof of 393.6K is less than 1000K, it can be monitored by a bolometer, and a thermocouple,

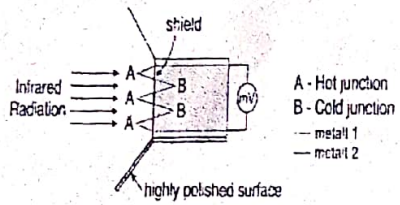


- P and Q are standard resistors while R is a variable resistor. In absence of infra-red radiation, the variable resistor is adjusted until the center-zero galvanometer, G shows no deflection.

- The value of resistance of R_O at room temperature is noted. The resistance of the blackened metal strip, $R_B = \frac{Q}{R_0}xP$ is obtained. Infrared radiation is now directed onto the metal strip.
- The experiment is repeated and the new value of the resistance R'_B is calculated.
- It is noted that $R'_B > R_B$, therefore the temperature can be monitored.

OR

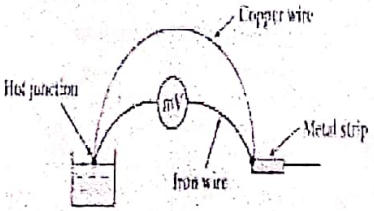
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- Heat radiations produced by blackened metal strip is directed onto a blackened junction A.
- The junction absorbs the radiation causing its temperature to rise above that of a junction B.

The temperature difference between A and B causes a thermometric emf to be generated. The emf generated is measured which deflects as a result and proportional to the temperature difference.

OR



- Two dissimilar metals are joined together via milli-voltmeter, one junction is attached to a thin blackened metal strip.
- The other junction is shielded from the infrared and its temperature is kept at a stable reference temperature.

- Infrared radiation is allowed to fall on the metal strip. The heat generated at the metal strip is absorbed and heats up the junction.
- The temperature difference between the two junctions generates emf between them.
- The generated emf is registered by the milli-voltmeter. This emf is proportional to the temperature difference.

ITEM 7:

The rapid strokes ensure that no heat escapes which implies that the process occurs adiabatically, since, the work is done on the gas, internal energy of the gas increases.

Internal energy increases with temperature accounts for the rise in temperature.

Work done on the gas:

$W = Cv\,\Delta T$

$350 = Cv \times (290 - 273)$

$Cv = 20.59\ J\ K^{-1}\ mol^{-1}$

Gas Constant R calculation:

$R = (PV)/T$

But $V = M/\rho = (32 \times 10^{-3})/1.43 = 0.022\ m^3$

$R = (1.01 \times 10^5 \times 0.022)/273 = 8.139\ J\ mol^{-1}\ K^{-1}$

$Cp = R + Cv = 8.139 + 20.59 = 28.729\ J\ K^{-1}\ mol^{-1}$

$\gamma = Cp/Cv = 28.729/20.59 = 1.40$

Adiabatic process pressure calculation:

$PV^{\gamma} = constant$

$P_2 = P_1(\frac{V_1}{V_2})^{\gamma} = 1.01 \times 10^5 \times (0.022/0.0166)^{1.4} = 1.50x10^5 Pa$

Specific heat capacity at constant pressure (Cp) and Ratio (γ):

Since $P_2 = 1.50x10^5\ Pa < 1.64x10^5 Pa$, then there is no power disruption

- The coloring of the edge of the text displayed is due to chromatic aberration. The light incident on the lens system is dispersed into the components of white light. Each coloured light is brought to its own focus. When these coloured lights overlap, the image of the text is coloured.
- The lens is placed in contact with a concave lens of different refractive indices to form an achromatic doublet. As the convex lens disperses white light, the concave lens cancels the dispersion. The image of the text will be free from the coloring.
- Sharp and clear focus is ensured by adjusting the position of focusing lens with respect to the source to be at conjugate foci; this allows the image of the source to get formed in the position of the lens and avoid the formation of the image of the source on the screen.

(b) The resultant displacement of the two waves superposing

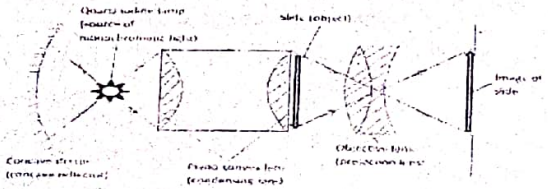
$$y = 0.1\sin(200\pi t - \frac{20\pi x}{17}) + 0.1\sin(200\pi t - \frac{20\pi x}{17})$$

$$y = 0.2\sin(\left(200\pi t - \frac{20\pi x}{17}\right)/2 + (200\pi t + \frac{20\pi x}{17})/2)\cos((200\pi t - \frac{20\pi x}{17} - (200\pi t + \frac{20\pi x}{17})/2) = 0.2\sin(200\pi t)\cos(-\frac{20\pi x}{17})$$

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$$y = 0.2\cos\frac{20\pi x}{17}\sin 200\pi t$$

For projection lens;



- A quartz iodine lamp (source of monochromatic light) is placed at the Centre of a concave reflector / mirror that ensures very high intensity source of light by reflecting back light rays that would otherwise be wasted at the back of the lamp thereby preventing light loss.
- The Plano convex lenses/condensing lenses are used to concentrate light rays onto the slide (object).
- The objective/ projection lens ensures that a magnified upright image of the slide is formed on the screen.
- The screen is where the real image of the slide is formed.

ITEM 8:

a) Sunlight consist of colored lights of varying wavelength in a given medium.

When light is incident onto a given glass prism, at an angle of incidence for the different refractive indices for the different colored lights. Infrared with wavelength longer than ultra-violet is least deviated. Therefore dispersion takes place.

Since ultra-violet passes through the prism symmetrically, $n = \frac{\sin(\frac{D_{min}+A}{2})}{\sin(\frac{A}{2})} = \frac{\sin i}{\sin\frac{A}{2}}$

$1.5 = \frac{\sin i}{\sin 30} \Rightarrow i = 48.6^0$ *But For infra-red light,*

$$1.48 = \frac{\sin i}{\sin r} = \frac{\sin 48.6}{\sin r_1} \Rightarrow r_1 = 30.45^0$$

$$r_2 = A - r_1 = 60 - 30.45 = 29.55^0$$

Refraction at the second face

$$1.48\sin 29.55 = \sin i_2 \Rightarrow i_2 = 46.88^0$$

$D = (i_1 - r_1) + (i_2 - r_2) = i_1 + i_2 - A = 48.6 + 46.88 - 60 = 35.48^0$

since $D > 10^0$, *the prism will not effectively disperse the sun light.*

compare $y = 0.02\sin(200\pi t - \frac{40\pi x}{17})$ *With;*

$y = a\sin(wt - kx)$

$w = 2\pi f = 200\pi$

$f = 100Hz$

$k = \frac{2\pi}{\lambda} = \frac{40\pi}{17}$

$\lambda = 0.85m$

$$v = f\lambda = 100x0.85 = 85ms^{-1}$$

Since, $v = 85ms^{-1}$ *is less than* $300ms^{-1}$, *the rodents will not be successfully repelled.*

b) Assuming no heat is lost to the surrounding;

Heat gained by oil and calorimeter= heat lost by the hot solid

$$m_o c_0(\theta_2 - \theta_1) + m_c c_c(\theta_2 - \theta_1) = m_s c_s(\theta_3 - \theta_2)$$

$$0.08c_o(40 - 15) + 0.120x409(40 - 15) = 0.200x\,450(100 - 40)$$

$$c_o = 2100Jkg^{-1}K^{-1}$$

Since C_0 *lies between* $1800Jkg^{-1}K^{-1}$ *and* $2200Jkg^{-1}K^{-1}$, *the cooling oil is of standard quality.*

Heat is lost during the transfer of the hot solid and some is absorbed by the insulating material. Continuous flow is the best since the heat capacity of the apparatus is not required, because;

- Readings are taken under steady states.
- The glass jacket is evacuated, there is no heat lost by convection and conduction.
- Heat loss to the surrounding is catered for by repeating the experiment,
- No cooling correction is needed.

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ITEM 10:

When the chemical fluid temperature is significantly reduced, the temperature of the coil is reduced!

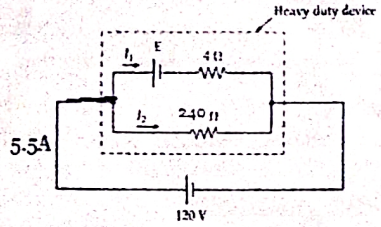
The internal energy, E, of its atoms is also reduced. The amplitude of vibration of the atoms reduces, meaning space for the free electrons increases. The number of electrons being transferred per second increases, so the current in the coil rapidly increases. This leads to too much heating that causes the coil to get burnt

$R_{373} = \frac{V^2}{P} = 12^2 / 36 = 4\ \Omega$

$\rho_{373} = \rho_{273}(1 + \alpha\vartheta) = 1.08 \times 10^{-6}(1 + 8 \times 10^{-3} \times 100) = 1.944 \times 10^{-6}\ \Omega m$

$l = (R_{373} \times A) / \rho_{373} = (4 \times 0.10 \times 10^{-6}) / (1.944 \times 10^{-6}) = 0.21\ m$

Since the length of the burnt coil matches the length of the available nichrome, it can be replaced with the new nichrome wire.



$240 \times I_2 = 120 \Rightarrow I_2 = 0.5\ A$

$I_1 + I_2 = 5.5$

$I_1 + 0.5 = 5.5 \Rightarrow I_1 = 5.0\ A$

$E = -4 \times I_1 + 240 \times I_2$

$= -4(5.0) + 240 \times 0.5 = 100V$

Hence E = 100 V

Since $I_1 = 5.0\ A > 4.5\ A$) and E is exactly 100 V, the device is able to provide a constant supply of the chemical.

b) Efficiency, η = (Back e.m.f. / Applied P.d.) × 100%

Efficiency, η = (100 / 120) × 100% = 83.3%

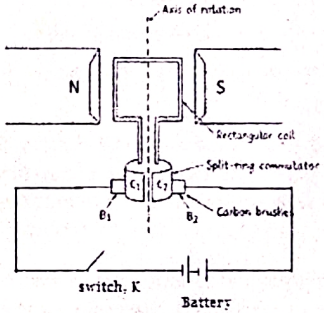
Since the efficiency of the device is less than 85%, it does not meet the target.

Efficiency Improvements of DC Motor

Efficiency of the d.c. motor is increased by increasing the back e.m.f. Since back e.m.f. is proportional to the speed of rotation, the following improvements can be made:

- Laminating the soft iron armature with insulators in order to reduce eddy currents. The magnetic forces which would oppose the rotation of the coil are reduced. This increases the speed of rotation.
- Lubricating the small gaps between the commutators and carbon brushes. This minimizes friction forces and work done against friction is minimized.
- Using a coil of low electrical resistance to reduce ohmic losses which would have caused the heating of the coil.

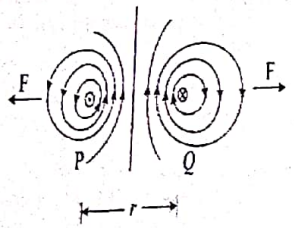
The heavy duty device is a d.c. motor



- When the d.c. supply is switched on, direct current flows through the coil. Each side of the coil experiences an equal but opposite magnetic force.
- A magnetic torque is exerted on the coil causing rotation of the coil about an axis perpendicular to the magnetic field, until the commutators lose their contact with the carbon brushes.
- The flow of current stops, but the inertia of the coil keeps it moving. When the commutators regain contact with the carbon brushes, the direction of current in the coil is reversed.
- The reversal of current enables constant magnetic torque in one direction. Thus, the chemical is continuously circulated.

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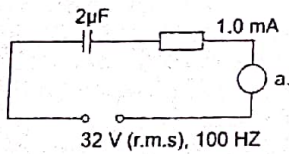
ITEM 11:



Both wires P and Q produce circular magnetic fields around them.

The magnetic field due to each wire interact and create a region of strong magnetic field between them pushes the wires towards a region of weak magnetic field. Therefore the wires experience repulsive force, F.

Force per metre, $F = \frac{\mu_0 I_1 I_2}{2\pi r} = 5.0x10^{-6}N \Rightarrow r = \frac{4\pi x10^{-7} x1.5x3.0}{2\pi(5.0x10^{-6})} = 0.18m$



The capacitive reactance, $X_C = \frac{1}{2\pi fC}$ is inversely proportional to the frequency of a.c source.

As the frequency increases, the capacitive reactance of the circuit reduces. The rate of charging and discharging of the capacitor increases. More charge flow per second leads to increased current flow. Therefore, the ac meter registers current rise from 1.0mA to 2.0 mA.

When a dc source is used, the capacitor charges and current flows as the capacitor is charging. When the capacitor is fully charged, charge flow in the circuit stops.

The meter registers zero current.

Total impedance $Z = \frac{V}{I} = \frac{32}{0.001} = 32000\ \Omega$

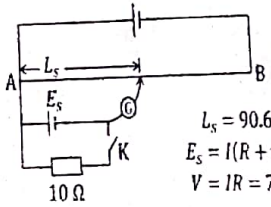
$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi x100x2.0x10^{-6}} = 795.8\ \Omega$

$Z = \sqrt{X_c^2 + R^2}$

$32000^2 = R^2 + 795.8^2$

$R = 31990\ \Omega$

Potentiometer Circuit & Internal Resistance



$L_s = 90.6$ cm

$E_s = I(R + r) = 90.6\,K$

$V = IR = 75.5\,K$

About internal Resistance

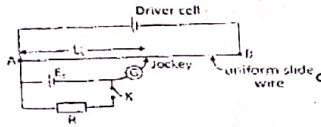
Given: Balance length with switch K open (E.M.F. balance length), l_1 = 90.6 cm

Balance length with switch K closed across resistance R = 10 Ω (Terminal p.d. balance length),

$l_2 = 75.5$ cm $\Rightarrow \frac{(10+r)}{10} = \frac{90.6}{75.5} \Rightarrow r = 2.0\Omega$

Since the internal resistance lies between 1.8 Ω and 2.2 Ω, the battery performs optimally.

b)



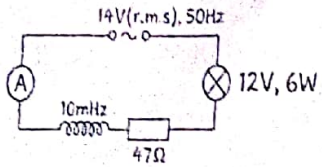
- The circuit is connected as shown in the diagram with driver cell, slide wire AB, switch K, jockey, and protective-resistor.
- With switch K open, the jockey is moved along the slide wire AB until the galvanometer G shows no-deflection.
- The balance length, l_1, is measured.
- Switch K is closed and the procedure is repeated to obtain the new balance length, l_2
- The internal resistance $r = R_0(\frac{l_1}{l_2} - 1)$ is calculated.

Precautions:

- The driver cell must provide a steady current to ensure consistent potential gradient along the wire.
- The jockey must not be scraped along the slide wire so that the wire maintains a uniform cross-section and potential drop per cm remains constant.
- A very sensitive galvanometer should be connected in series with a protective resistor to reduce initial current and avoid damage/over damping.

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ITEM 12:



The resistance of the cabin bulb, R_B

$R_B = \frac{V^2}{P} = \frac{12^2}{6} = 24\Omega$

Total Resistance, $R_T = 24 + 47 = 71\Omega$

Inductive reactance $X_L = 2\pi fL = 2\ \pi x50x10x10^{-3} = 3.14\Omega$

Impedance, $Z = \sqrt{R^2 + X_L^2} = \sqrt{71^2 + 3.14^2} = 71.1\Omega$

The current supplied by the circuit,

$I = \frac{V}{Z} = \frac{14}{71.1} = 0.195A$

Since I is less than 0.4 A, the current detector's coil will not get burnt

- When switch, S_1 is closed it allows the flow of alternating current through the circuit. The changing current induces back emf in the inductor. This back emf continuously opposes the applied current. The resulting current in the circuit becomes very small and makes the bulb to dim.
- The magnetic torque exerted on the moving coil of the current detector keeps changing its direction due to changing current. The pointer vibrates at a frequency of a.c and makes it difficult to measure current.
- When switch S_2 was closed it created an alternative route for a.c. in the first half of a.c when P is positive with respected to Q, device D is forward biased and allows current to flow through it.
- The back emf in the coil is zero initially, more current flows through the circuit and the bulb immediately attains full brightness. In the next half cycle of a.c , the device D is backward biased, and blocks current but since the frequency of the a.c is high
- As current flows through the inductor, the back emf induced opposes the growth of current in the coil, the dimming of the bulb is not realized.

The operating current for the bulb, $I = \frac{P}{V} = \frac{6}{12} = 0.5A$

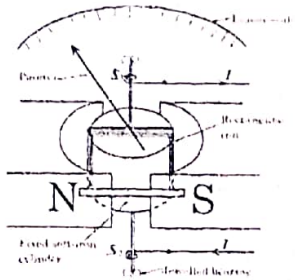
The supplied current, $I_2 = \frac{E}{R}$; since back emf is zero.

Therefore, $I_2 = \frac{12}{71.0} = 0.169\ A$ Since,

I_2 is less than 0.169 A, the bulb will not blow and it does not require a shunt.

b) When the d.c source is switched on, the current rapidly increases to maximum. During the increase, back emf is induced in the coil, it opposes the applied current and the bulb instantly dims.

When the current attains a steady maximum value, the back emf in the coil becomes zero. The bulb glows to full brightness. Since d.c flows in one direction, the current detector measures the current since the magnetic torque on the coil of the current detector acts in one direction.



- The current enters the coil through the lower hair spring and leaves the coil through the upper hair spring.
- The coil experiences an equal but opposite magnetic force on each side of the coil, These forces constitute of a magnetic torque τ=BINA.
- The coil together with the pointer rotates over linear scale until it is stopped by the restoring torque, $\tau_r = c\theta$
- Therefore, $I = \frac{c\theta}{BAN}$; Implying, $I \propto \theta$, *hence linear scale.*

ITEM 13:

For the circuit to pick the radio signals, the frequency of the tuning circuit must be in electrical resonance with the frequency of the of the radio signal.

At resonance, the current flow in the circuit is maximum when the impedance is minimum. The capacitive reactance is equal to inductive reactance i.e $X_L = X_C$

Implying that; $2\pi fL = \frac{1}{2\pi fC}$

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But $C_0 = \frac{\varepsilon_0 A}{d} = \frac{1}{0.5 \times 10^{-3}} x 8.85x10^{-12} x 1.5x10^{-3} = 2.655x\ 10^{-11} F$

For capacitors in parallel, $C = 10C_0 = 10\ x\ 2.655x10^{-11} = 2.655x10^{-10} F$

$$L = \frac{1}{4\pi^2 f^2 C} = \frac{1}{4\pi^2 x(62x10^6) x\ 2.655x10^{-10}} = 2.4x10^{-8} H$$

The magnetic flux density, $B = \mu_0 nI$ But; $n = \frac{200}{l}$

Therefore; $l = \frac{B}{\mu_0 I} = \frac{1.0 \times 10^{-3}}{4\pi x10^{-7} x0.2} = 0.05m$

When the current is in forward bias, electrons move from the n-type region into the holes of p-type region.

The increased density of electrons and holes near the junction leads to their recombination. This creates photons equal to the energy lost by the electrons when falling from conduction band to valence band.

The energy of the radiation is a characteristic of wavelength and frequency which determine the color. At the lower frequency, the wave length is longer which corresponds to red light.

Yellow light has an intermediate wavelength followed by green light with shorter wavelength.

A charged capacitor discharges through the coil. The flowing current produces magnetic field around the inductor and energy is stored in the magnetic field.

When the capacitor is completely discharged, the magnetic field of the coil collapses. As it collapses an emf is induced across the terminals of the coil which recharges the capacitor in opposite direction. This continuous charging and discharging ensures continuous flow of charge in the circuit which accounts for the continuous flow of current.

The a.c meter therefore, registers the root mean square value of current in the circuit which is constant.

b) When a solid piece of soft iron is inserted into the coil, the self-inductance increases, and due to change of magnetic flux, a large p.d is induced in a soft

iron. Eddy currents are created within the soft iron. These eddy currents create internal energy which is realized as heat.

Eddy currents are useful in an induction furnace to melt metals and separate impure materials from pure metals.

A metal to be melted is put in a crucible tank and the crucible tank is surrounded with a solenoid.

The solenoid is connected to a high frequency a.c source. As the current flows, it creates a very high changing magnetic flux which induces eddy currents in these parts that are conducting. The internal energy of the conducting parts of the metals rises and melt the metal.

The melted metal flow to one part of the crucible leaving the un-melted impurities on the other part. When the melted metal cools, it is removed when it is pure.

ITEM 14:

Protecting the galvanometer

Given N=250 turns, A =$4.0x10^{-4}$, $B = 0.025T$, $k = 3.0\ X10^{-5} Nmrad^{-1}$, $\theta = 0.4\ rad$

Torque of the coil, $\tau = BINA$

At equilibrium: $BINA = k\theta \Longrightarrow I = \frac{k\theta}{BNA} = \frac{3X10^{-5}X0.4}{0.025X250\ X\ 4X10^{-4}} = 4.8\ x\ 10^{-3}\ A$

(Current corresponding to the deflection of 0.4 rad)

Resistance of protective wire $\Longrightarrow R = \frac{\rho l}{A} = \frac{4.9x10^{-8} x\ 0.5}{4.25x10^{-10}} = 57.6471\Omega$

Power dissipated in the wire to be used to protect the galvanometer and the protective resistor.

Total resistance, $R_T = 57.6471 + 2 = 59.6\ \Omega$

$$P = I^2R = (4.8\ X10^{-3})^2 X59.6 = 1.373x10^{-3} W$$

Electrical energy, $E = Pxt = 1.373x10^{-3} x\ 30 = 4.12x10^{-2} J$

Since $4.12x10^{-2} J < 4.55X10^{-2} J$, *the hair springs will not burn out.*

For the radio tuning circuit:

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Inductance of solenoid; $L = \frac{\mu_0 N^2 A}{l} \Rightarrow C = \frac{l}{fV} = \frac{0.0048}{50X30} = 3.2X10^{-6}F$

But; $A = \pi r^2 = \pi(0.0142)^2 = 6.33x10^{-4}m^2$

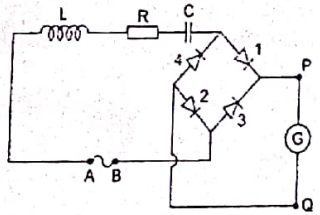
$$L = \frac{4\pi x10^{-7} x600^2 x6.33x10^{-4}}{0.227} = 1.26x10^{-3}H$$

Required capacitor for 2.6 kHz

Resonance occur when; $f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1.26x10^{-3}x3.2x10^{-6}}} = 2506.5\ Hz.$

The moving coil galvanometer, measures d.c but not a.c. when a.c is switched on the direction of current through the galvanometer keeps on changing repeatedly. The pointer therefore vibrates rapidly about the equilibrium at the frequency of the source, seeming not to deflect. This makes it difficult to measure alternating current.

b) Modification



- When the circuit is switched on and A is positive, with respect to B in the first half of a.c cycle, diodes 1 and 2 are forward biased and allow current flow while diodes 3 and 4 blocks the flow of current .
- In the next half cycle of ac, when B is positive with respect to A, only diodes 3 and 4 are forward biased and allow current to flow while diodes 1 and 2 block the flow of current.
- In both a.c cycles, current flows through the galvanometer in one direction from P to Q. This makes the galvanometer measure the root mean square value of the current.

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi x40x3.2x10^{-6}} = 1243.4\ \Omega$$

$$X_L = 2\pi fL = 2\pi x40x1.26x10^{-3} = 0.3167\Omega$$

$$Z = \sqrt{(X_C - X_L)^2 + R^2}$$

$Z = \sqrt{(1243.4 - 0.317)^2 + 59.6^2} = 1244.5\Omega$

The current, $I = \frac{V}{Z} = \frac{12}{1244.5} = 0.0096A$

ITEM 15:

Selecting a suitable transformer

Given; primary voltage, $V_p = 132kV = 132000V$

Secondary voltage, $V_s = 11kV = 11000V$

Input current, $I_P = 500A$

Minimum efficiency, $\eta = 95\%$

Maximum allocation power loss= 1.32MW

Power input, $P_{in} = I_p V_p = 132000\ x500 = 66MW$

Minimum efficiency requirement

Power loss at 95% efficiency is

$$P_{loss} = (1 - \eta)P_{in} = (1 - 0.95)x\ 66 = 3.3\ MW$$

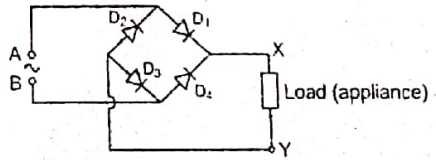
The power loss is 3.3 MW which exceeds the permitted loss of 1.32 MW. Therefore, a transformer operating at exactly 95% efficiency is not suitable.

Minimum efficiency require:

$$\eta = \left(\frac{P_{in} - P_{loss}}{P_{in}}\right) x\ 100\% = \left(\frac{66 - 1.32}{66}\right) x\ 100\% = 98\%$$

Therefore; the manager should purchase a transformer with an efficiency of at least 98% under the conditions stated for operation.

To operate the 100V d.c appliance from an a.c supply; the manager should use a a bridge rectifier bridge circuit to convert a.c to d.c



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- During the first half cycle, terminal A is positive while B is negative, Diodes D_1 *and* D_3 are forward biased and conduct while diodes D_2 *and* D_4 are reverse biased and don't conduct.
- Current flows in direction X to Y via the load
- During the second half cycle, terminal B becomes positive while A becomes negative, Diodes D_2 *annd* D_4
- Are forward biased and conduct while diodes D_1 *and* D_3 are reverse biased and don't conduct.
- Current again flows in the direction X to Y via the load
- Since the current flow through the load is in the same direction, then it is said to be direct current (d.c)

Determining the capacitance for resonance:

Given inductance; L= 46.4 mH, R=10Ω, peak voltage, $V_0 = 10V$, $I_{rms} = 0.4\ A$, $f = 50Hz$.

$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{10}{\sqrt{2}} = 7.07V \Rightarrow$ *Impedance,* $Z = \frac{V_{rms}}{I_{rms}} = \frac{7.07}{0.4} = 17.68\ \Omega$

The inductive reactance:

For LCR circuit: $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$(X_L\text{-}X_C) = \sqrt{17.68^2 - 10^2} = 14.58\ \Omega$

At resonance, $X_L = X_C$, *so the impedance* $(Z = R) = 10\Omega$, *which is disagrees with the given measurement.*

Capacitor required for resonance

At resonance, $f_0 = \frac{1}{2\pi\sqrt{LC}}$

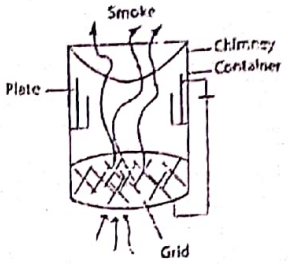
Therefore, $C = \frac{1}{4\pi^2 f_0^2 L} = \frac{1}{4\pi^2 x 50^2 x 46.4\ 10^{-3}} = 218\ \mu F$

A capacitor of 218μF is required for resonance at 50Hz to occur if the stated inductance is correct.

b) The residents' concerns and treating of the factory fumes.

Untreated fumes are dangerous as they may contain harmful gases like carbon dioxide which reduces blood ability to transport Oxygen and other smoke particles that aggravate Asthma and other respiratory disorders.

Therefore, the factory fumes can be treated using an electrostatic precipitator.



- ✓ In a chimney, there is a grid which is negatively charged with respect to the plates.
- ✓ As smoke passes through the grid, its particles are negatively charged
- ✓ When the smoke reaches the regions of plates, the particles are attracted to the plates
- ✓ As a result, smoke gets out of the chimney free of particles.
- ✓ The particles accumulating on the palates fall off in the container when they become heavy enough
- ✓ Thereby reducing particulate pollution which protects the health of the nearby residents.

THE END

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